Wisdom Favour Chizoba

Manufacturing Systems & Digital Transformation Engineer

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PROFESSIONAL EXPERIENCE

Oct. 2026 – present

System Developer in Manufacturing (Working Student)

Schiederwerk GmbH, Nuremberg

- Development and automation of production and quality reporting using Grafana, MariaDB, SQL, n8n, and Scriptcase.
- Integration and evaluation of production data from MES/ERPs to create end-to-end transparency along the manufacturing process.
- Development and enhancement of manufacturing execution and production information systems to support manufacturing, quality assurance, and traceability.
- Collaborated with production, quality assurance, process engineering, and IT in the digitalization and continuous improvement of manufacturing processes.

Oct. 2025 – Sept. 2026

Customer Quality Engineering (Working Student)

ams OSRAM, Regensburg

- Produced 100+ customer-facing reliability monitoring reports and CDC documentation packages for semiconductor and LED product families, consolidating qualification, reliability, and test data to support customer quality and reliability requirements.
- Supported PPAP and quality documentation activities through structured capture, validation, and consolidation of reliability and qualification data.
- Analyzed reliability test results (HTOL, WHTOL, PTC, TC, T&HB, DEW, CDM, and corrosion testing) to identify failure trends, assess root causes, and support customer quality reporting.
- Collaborated with Customer Quality, Production Quality Engineering, Reliability Engineering, and Manufacturing across global sites to resolve quality issues.

Apr. 2025 – Oct. 2025

Quality Systems Engineering (Bachelor's Thesis)

Mubea Aerostructures GmbH, Munich

- Owned the end-to-end design and implementation of a Microsoft Power Apps solution that transformed Root Cause Analysis (8D, 5 Why, Ishikawa) from a manual process into a standardized digital workflow.
- Established a structured foundation for quality data by converting unstructured RCA documentation into traceable, analyzable datasets.
- Improved audit readiness and supported compliance with EN9100 and EASA Part-145/Part-21 through automated documentation and standardized workflows.
- Optimized quality processes and reduced documentation inefficiencies in aerospace manufacturing.

Apr. 2024 – Oct. 2025

Quality Engineering (Working Student)

Mubea Aerostructures GmbH, Munich

- Supported process planning and quality assurance through data-driven analysis to ensure production performance and process stability.
- Conducted Gemba walks on the shop floor to verify adherence to quality standards (5S, Standard Work).
- Contributed to the digitalization of quality documentation and improved data availability.
- Supported First Article Inspections (FAIs), customer complaints and NCRs, and CAPA processes.

Oct. 2023 – Apr. 2024

Data Engineering (Internship)

ZF Friedrichshafen AG, Friedrichshafen

- Developed management dashboards in Power BI to provide visibility into ZF Supplier Innovations, from submission through implementation, including financial savings.
- Built data pipelines by cleaning and transforming supplier and process data using Python, Excel, VBA, and SQL, and connected database APIs to Power BI.
- Delivered monthly and ad hoc analytical insights to support management decisions.
- Reduced manual reporting effort by automating data pipelines and KPI updates.

PROJECTS

AnyLogic Digital Twin for Warehouse Sequencing & Downtime Minimization

Multi-method simulation with Java agents for dynamic sequencing between high-bay storage and body shop; provides an analytical basis for weighing storage costs against the financial risk of production downtime.

Technologies: AnyLogic · Java

AnyLogic Simulation for Quality Inspection Station Capacity Optimization

Stochastic discrete-event simulation with parameter variation experiments and 95% confidence intervals; demonstrated a 71% reduction in mean cycle time (16.0 → 4.7 min) from adding inspection staff, outperforming an equipment upgrade.
Technologies: AnyLogic · Statistical Analysis

Discrete-Event Digital Twin for Quality Control & Bottleneck Analysis
Python/SimPy simulation of a multi-station production line with stochastic machine breakdowns (MTBF/MTTR), automated rework/scrap logic, and a real-time KPI pipeline (OEE, throughput, bottleneck detection).
Technologies: Python · SimPy

Industrial PLC-to-Cloud OEE Monitor
End-to-end IT/OT architecture: PLC control logic (IEC 61131-3) connected via Modbus TCP to a Python telemetry pipeline and a live-updating OEE dashboard; resolved scan-cycle overruns and counter drift in production.
Technologies: OpenPLC · Python · SQLite · Plotly Dash

Additional projects available at: https://chizobawisdom.github.io/Portfolio/index.html

EDUCATION

2027 (ongoing) **M.Eng. Applied AI for Digital Production Management**
Technische Hochschule Deggendorf

2025 **B.Eng. Industrial Engineering and Management**
Technische Hochschule Deggendorf

SKILLS

Industrial Data & AI
- Data analytics & machine learning with Python (pandas, NumPy, scikit-learn), SQL, and predictive modeling for anomaly detection, classification, and forecasting
- Design and implementation of key metrics such as First Pass Yield (FPY), defect rate, throughput, Work-in-Progress (WIP), lead time, and equipment utilization
- Industrial computer vision with OpenCV, TensorFlow, CVAT, and transfer learning for defect detection and image classification
- Data engineering & BI: ETL workflows, data models, KPI pipelines, dashboards, and automated reporting with Python, SQL, Power BI, and Excel

Manufacturing Systems, Quality & Reliability
- Digital twins & simulation with AnyLogic, SimPy, and MATLAB for evaluating queuing behavior, capacity limits, and downtime
- Development of solutions for tracking products, production steps, and quality events within complex manufacturing environments
- Quality management & continuous improvement: reliability monitoring, PPAP, APQP, RCA, CAPA, 8D, 5 Why, Ishikawa, SPC per IATF 16949, ISO 9001, EN9100

Automation, Integration & Digitalization
- Industrial automation: PLC concepts, Modbus TCP, OPC-UA, MES concepts, real-time monitoring in IT/OT architectures
- Digitalization of manufacturing processes with Power Apps, Power Automate, and VBA to standardize workflows
- System integration: connecting production systems, databases, APIs, and visualization platforms
- Engineering toolchain: Grafana, MariaDB, SQL, Python, Scriptcase, n8n, Power BI, SAP, SharePoint, Excel, KNIME, MATLAB, and modern web technologies

LANGUAGES

- German – B2 (telc certified)
- English – C2 (native speaker)